

SmolVLA RL Training

Reinforcement Learning for robotic rovers using Vision-Language-Action models.

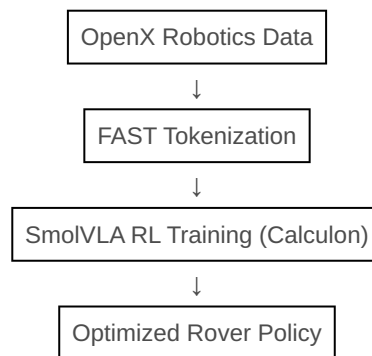
WHAT & WHY

- Integrates SmolVLA with Reinforcement Learning (RECAP).
- Enables high-resolution vision-language models for robot control.
- Addresses data efficiency and generalizability in robotics.
- Co-trains VLM backbones on large-scale robotics datasets.

HOW

- Uses FAST tokenizer for efficient large-scale data processing.
- Implements distributional RL (C51) for the VLM critic.
- Leverages PyTorch Accelerate for distributed multi-GPU training.
- Utilizes SmolVLM2-500M as the core pre-trained backbone.

OVERVIEW



KEY RESULTS

Dataset Scale: Trained on full OpenX datasets.

Model: Integrated SmolVLM2-500M backbone.

Architecture: C51 distributional critic implemented.

Input: Support for multi-camera video observations.

CLUSTER UNLOCK

Calculon storage and compute allowed training on the massive OpenX datasets without sampling or data reduction.

RESOURCES

Partition	Calculon GPU
Hardware	NVIDIA H200/A100
Storage	High-speed SSD
Software	PyTorch, Accelerate

LINKS

Repo	fhnw/rover/smolvla-rl
HF	IPEC-COMMUNITY/openx-lerobot

